

# Joint User Association and Power Allocation for Hybrid VLC/RF Networks

Siqi Li, Gang Chuai, Weidong Gao, KaiSa Zhang

Key Laboratory of Universal Wireless Communication, Ministry of Education  
Beijing University of Posts and Telecommunications, Beijing, 100876, China  
Email: {lisiqi2020, Chuai, gaoweidong, kaisa}@bupt.edu.cn

**Abstract**—In this paper, we study the optimization problem of joint user association and power allocation in visible light communication (VLC) and radio frequency (RF) heterogeneous networks. We design unsupervised neural network learning algorithms to solve this convex optimization problem to maximize the objective function. We compare the algorithm proposed in this paper with user association based on minimum distance and signal to interference plus noise ratio (SINR) and show that the scheme proposed in this paper can achieve better network performance.

**Index Terms**—hybrid networks, VLC, DNN, power allocation, user association

## I. INTRODUCTION

In the future network, VLC may become a beneficial complementary technology to indoor wireless communication technology. Compared with RF, VLC provides ultra-high bandwidth, zero electromagnetic interference, and free and abundant spectrum resources. Despite the obvious advantages of VLC, considering VLC network alone also has disadvantages such as small coverage area and easy to be obscured. In order to realize the construction of indoor high-speed visible optical networks, various indoor hybrid networking schemes have been tried. The hybrid VLC/RF heterogeneous network system can provide high speed connectivity while achieving seamless and reliable coverage [1].

[2] studied the power and bandwidth allocation problem in one RF access point (AP) and one VLC AP scenario, without considering the multi-VLC APs case and interference. [3] studied the problem of user association based on minimum distance of users in hybrid VLC/RF networks. In [4], [5], the resource allocation problem is solved based on convex optimization and reinforcement learning approaches, respectively. [6] proposed an iterative reshuffle based algorithm that asynchronously solves the joint optimization problem of user association and bandwidth allocation. All of these literatures have only studied the problem of user association or resource allocation individually or asynchronously.

In this paper, we study the problem of resource allocation in VLC/RF heterogeneous networks. We propose an unsupervised DNN algorithm to establish an objective function to maximize the network throughput with the aim of solving the user association and power allocation problems simultaneously. It can reduce the time complexity and improve the network performance compared with other algorithms.

## II. SYSTEM AND CHANNEL MODEL

In general, it is considered that RF APs can cover all areas of the room, while the communication range of VLC APs is related to their effective lighting area. To achieve full coverage of indoor optical network, it is usually necessary to arrange multiple VLC APs and one RF AP in a reasonable way. Fig 1 shows a heterogeneous network system model of VLC and RF in indoor environment. Due to the difficulty of implementing visible uplink, only the downlink is considered in this paper. As two kinds of wireless communication technologies, VLC

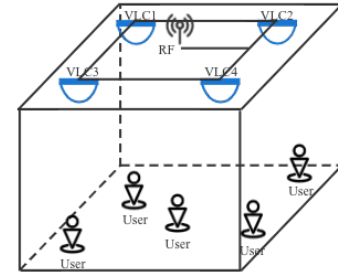


Fig. 1: System model.

and RF have significant differences in terms of bandwidth, channel model and other parameters, which are analyzed separately below.

### A. VLC Channel

Lambert radiation model was used as the light propagation model [2], and only the LOS channel is considered in this paper.  $h_{k,i}^{(v)}$  denotes the channel gain from VLC AP  $k$  to the  $i$ th user terminal, which can be defined as

$$h_{k,i}^{(v)} = \frac{(m+1)A_p}{2\pi d_{k,i}^2} \cos^m(\phi_{k,i}) g_f g_c(\psi_{k,i}) \cos(\psi_{k,i}) \quad (1)$$

where  $m$  is Lambert radiation index, which is related to the LED half-power emission angle,  $A_p$  is the photo diode (PD) detector receiving area,  $d_{k,i}$  is the distance between the transmitter and receiver,  $\phi_{k,i}$  is the angle of incidence of the receiver,  $\psi_{k,i}$  is the angle of radiance of the AP,  $g_f$  is the optical filter gain, and  $g_c$  is the light concentrator gain.

In this paper, all VLC APs are frequency division multiplexed, and users located in the VLC coverage overlap area will be subject to co-channel interference from the surrounding optical cells, which is considered as interference noise. At the

receiving end, the SINR value for user  $i$  with fixed access to VLC AP  $k$  is

$$SINR_{k,i}^{(v)} = \frac{(\rho \sqrt{P_{k,i}^{(v)}} h_{k,i}^{(v)})^2}{(B_{max}^{(v)}/N_k)N_0^{(v)} + \rho^2 \sum_{l=1, l \neq k}^{N_p} (\sqrt{P_{l,i}^{(v)}} h_{l,i}^{(v)})^2} \quad (2)$$

where  $\rho$  is the PD optical conversion constant,  $P^{(v)}$  denotes the transmit power of the access point with which the user establishes a communication link,  $B_{max}^{(v)}$  is the bandwidth of the VLC system, and  $N_0^{(v)}$  is the noise power spectral density of the VLC channel. Assuming that the AP allocates the available bandwidth equally to the users, when there are multiple users in a single AP, the rate that the network can provide is related to the number of users  $N_k$  accessed. The transmission rate available to user  $i$  by the VLC AP can be calculated from the Shannon-Hartley formula,

$$R_{k,i}^{(v)} = (B_{max}^{(v)}/N_k) \log_2(1 + SINR_{k,i}^{(v)}) \quad (3)$$

### B. RF Channel

The channel gain from the RF AP to the  $i$ th user is

$$h_{k,i}^{(r)} = h_0 \sqrt{10^{\frac{-L(d_{k,i})}{10}}} \quad (4)$$

The RF AP is considered to have a Rayleigh fading distribution in an indoor environment, and the propagation path loss is defined as  $L(d_{k,i})$ . In this paper, only indoor single RF coverage scenario is considered, there is no identical transmission channel, so the interference term is zero and SINR degenerates to SNR,

$$SNR_{k,i}^{(r)} = \frac{(\sqrt{P_{k,i}^{(r)}} h_{k,i}^{(r)})^2}{(B_{max}^{(r)}/N_k)N_0^{(r)}} \quad (5)$$

The transmission rate available to user  $i$  by RF AP is

$$R_{k,i}^{(r)} = (B_{max}^{(r)}/N_k) \log_2(1 + SNR_{k,i}^{(r)}) \quad (6)$$

### III. PROBLEM FORMULATION

In this section, we establish a joint optimization problem with the objective of solving the problem of user association and power allocation in a VLC/RF heterogeneous network, i.e., which AP should user  $i$  connect to and how much power is allocated to the  $N$  users connected to AP  $k$ .

We assume that each user is connected to only one AP. We use the throughput that can be achieved at the user end as a measure of network merit. Maximize the sum rate within the coverage area by linking and optimizing. Thus, the optimization problem that maximizes the rate can be mathematically formulated as

$$\mathcal{P} : \max \sum_{k=0}^K \sum_{i=1}^M a_{k,i} R_{k,i} \quad (7)$$

which subject to the following constraints:

$$\mathcal{C}_1 : \sum_{k=1}^K a_{k,i} = 1 \quad \forall i \in M \quad (8)$$

$$\mathcal{C}_2 : \sum_{i=1}^M P_{k,i} = P_{k,max} \quad (9)$$

$$\mathcal{C}_3 : P_{k,i} \geq 0 \quad (10)$$

where  $P_{k,max}$  is the total power that can be allocated to an AP. The constraint in (9)(10) illustrates that the sum of the power assigned to the users associated with access point  $k$  cannot exceed  $P_{k,max}$  and must be greater than zero.

The optimization problem (7) is non-convex and it is difficult to obtain the optimal strategy by traditional methods. Deep learning algorithms are used to reduce the computational complexity through matrix operations. Neural networks can be used to learn their input-output relationships and thus simulate highly nonlinear and complex systems.

### IV. USER ASSOCIATION AND POWER ALLOCATION BASED ON DNN

We propose an unsupervised deep learning neural network(DNN) based learning algorithm to approximate the optimal value of user association and power allocation to solve the non-convex optimization problem of the objective equation. The DNN model design is shown in Fig 2. The fully connected

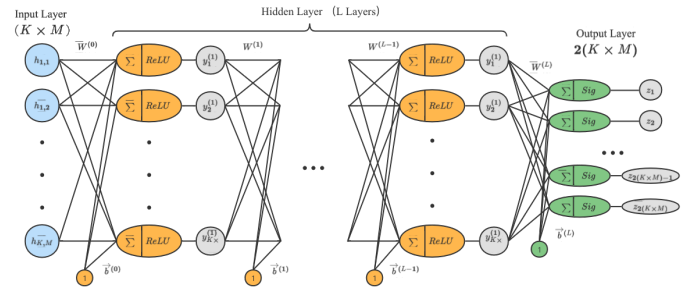


Fig. 2: DNN model.

neural network we designed contains one input layer, multiple hidden layers and one output layer. The channel gain vectors between  $K$  APs and  $M$  users are used as input data.  $W$  and  $\vec{b}$  are the weight matrix and bias vector, respectively. The forward computation of each layer of DNN network is divided into two steps: linear forward and activation forward. After linear forward, the output of the  $l$  layer is

$$\vec{y}^{(l)} = W^{(l-1)} \vec{y}^{(l-1)} + \vec{b}^{(l-1)} \quad (11)$$

After activation forward, the output of  $l$  layer is

$$\vec{y}^{(l)} = g(\vec{y}^{(l)}) \quad (12)$$

Here activation forward uses  $g(x)$ , which is some nonlinear activation function. In this paper, the hidden layer uses ReLU, i.e.  $g(x)_{ReLU} = \max(0, x)$ , and the output layer uses Sigmoid, i.e.  $g(x)_{Sig} = \frac{1}{1+e^{-x}}$ , to map the variables to between  $[0, 1]$ . The output layer is set up with  $2(K+M)$  nodes which can be expressed as,

$$\vec{z} = g_{Sig}(W^{(l)} \vec{y}^{(l)} + \vec{b}^{(l)}) \quad (13)$$

According to the output vector  $\vec{z}$ , the optimization objectives, i.e. the user association vector  $\vec{a}$  and the power allocation vector  $\vec{p}$ , can be expressed as,

$$\vec{a} = \max(\vec{z}_{[1:K \times M]}) \quad (14)$$

$$\vec{p} = \frac{\vec{a} \cdot \vec{z}_{[K \times M+1, 2(K \times M)]}}{\sum \vec{a} \cdot \vec{z}_{[K \times M+1, 2(K \times M)]}} \cdot P_{max} \quad (15)$$

The first  $K \times M$  nodes of the output layer are used to obtain the results of user association and the last  $K \times M$  nodes to obtain the results of power allocation.

In order to find the optimal value of user association and power allocation, it is important to determine an appropriate loss function to learn the weight matrix and bias vector for a given input data. Usually, the loss function is defined as a negative objective function, i.e.,  $Loss(\vec{a}, \vec{p}) = -R(\vec{a}, \vec{p})$ .

## V. SIMULATION RESULTS

Consider an indoor wireless heterogeneous network with both VLC and RF network coverage, where there is one RF AP arranged in the center of the room and four VLC APs distributed around the room. The users are randomly distributed in any location of the room. Table I shows the performance parameters used to evaluate the performance of the proposed algorithm.

TABLE I: Simulation Parameters.

|     | Name                   | Value      | Name            | Value               |
|-----|------------------------|------------|-----------------|---------------------|
| RF  | $L(d_0)$               | 68dB       | $N_0^{(v)}$     | $10^{-19} A^2 / Hz$ |
|     | $B_{RF}$               | 20MHz      | $P_{RF}$        | 2W                  |
|     | $A_p$                  | $0.01 m^2$ | $m$             | 1                   |
| VLC | $g_f$                  | 1          | $g_c$           | 2.25                |
|     | $\rho$                 | 1          | $N_0^{(v)}$     | $10^{-24} A^2 / Hz$ |
|     | $P_{VLC}$              | 5W         | $B_{VLC}$       | 100MHz              |
|     | Learning rate          | 0.001      | Mini batch size | 100                 |
| DNN | Number of hidden layer | 5          |                 |                     |

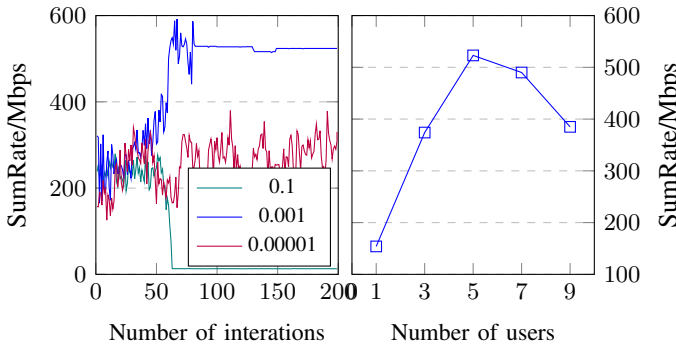


Fig. 3: System sum rate versus number of iterations.

Fig 3 shows that the neural network converges better for a learning rate of about 0.001 in a room of 10m×10m with 5 users. Higher learning rate will result in overfitting, and lower learning rate will result in slow convergence and difficult convergence. Traditional algorithms using minimum distance and maximum SINR for user association while sharing power

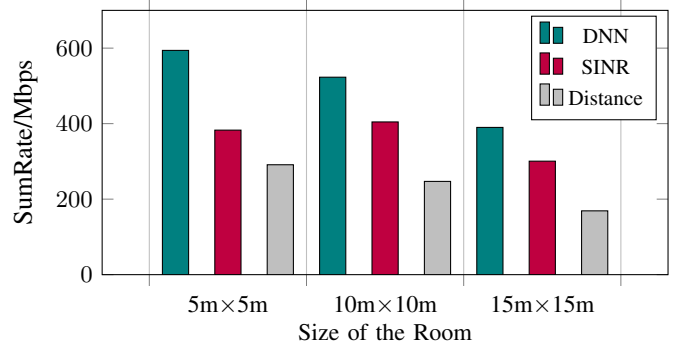


Fig. 5: System sum rate versus size of the room.

on average are used as comparison algorithms. Fig 5 shows the sum rate of the heterogeneous network that can be achieved by different algorithms in different sizes of rooms, and the best results are achieved by the joint solution algorithms proposed in this paper. It can also be seen that the larger the room, the further the distance from the AP access point, the smaller the sum rate can be achieved. Fig 4 shows that as the number of users grows the sum rate can be achieved by the system saturates, and when the number of users continues to increase, the sum rate decreases due to increased interference. At this point, if we want to increase the sum rate of the system, we need to increase the number of APs.

## VI. CONCLUSION

In this paper, we build a convex optimization problem for user association and power allocation in VLC/RF networks and solve it using an unsupervised DNN algorithm. We finally verify the optimal performance of the algorithm proposed in this paper. We also verify the variation pattern of the system sum rate as the room size and the number of users vary. In the future, we will continue our research, including such problems solved by using reinforcement learning, etc.

## REFERENCES

- [1] M. Ayyash et al., "Coexistence of WiFi and LiFi toward 5G: concepts, opportunities, and challenges," in IEEE Communications Magazine, vol. 54, no. 2, pp. 64-71, February 2016.
- [2] M. Kashef, M. Ismail, M. Abdallah, K. A. Qaraqe and E. Serpedin, "Energy Efficient Resource Allocation for Mixed RF/VLC Heterogeneous Wireless Networks," in IEEE Journal on Selected Areas in Communications, vol. 34, no. 4, pp. 883-893, April 2016.
- [3] I. Stefan and H. Haas, "Hybrid Visible Light and Radio Frequency Communication Systems," 2014 IEEE 80th Vehicular Technology Conference (VTC2014-Fall), 2014.
- [4] M. Obeed, A. M. Salhab, S. A. Zummo and M. -S. Alouini, "Joint Load Balancing and Power Allocation for Hybrid VLC/RF Networks," GLOBECOM 2017 - 2017 IEEE Global Communications Conference, 2017.
- [5] B. S. Ciftler, M. Abdallah, A. Alwarafy and M. Hamdi, "DQN-Based Multi-User Power Allocation for Hybrid RF/VLC Networks," ICC 2021 - IEEE International Conference on Communications, 2021.
- [6] A. Gupta, A. Banerjee, K. Pathak and S. Srivastava, "On Association and Bandwidth Allocation for Hybrid RF/VLC Systems," 2018 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), 2018.